

Day 3: Introduction to IP Addressing - Answers

Topic: IPv4 and IPv6

1. Key Differences: IPv4 vs IPv6

IPv4	IPv6
32-bit address length	128-bit address length
Written in dotted decimal, e.g. 192.168.1.1	Written in hexadecimal separated by colons, e.g. 2001:db8::1
About 4.3 billion addresses (2^{32}); exhausted	About 3.4×10^{38} addresses (2^{128}); practically unlimited
Header is variable (20-60 bytes) and includes a checksum	Fixed 40-byte header, simpler, no checksum
Uses NAT widely to save addresses	NAT not needed; every device can have a global address
Address configuration: manual or DHCP	Supports SLAAC (auto-configuration) as well as DHCPv6
IPsec is optional	IPsec support is built in (originally mandatory)
Uses broadcast	No broadcast; uses multicast and anycast

2. Static IPv4 Assignment (192.168.10.0/24)

Network: 192.168.10.0/24, Subnet mask: 255.255.255.0. The router is 192.168.10.1 and acts as the default gateway, so the devices use the addresses after it.

Device	IP Address	Subnet Mask	Default Gateway
Laptop	192.168.10.10	255.255.255.0	192.168.10.1
Phone	192.168.10.11	255.255.255.0	192.168.10.1
Smart TV	192.168.10.12	255.255.255.0	192.168.10.1
Printer	192.168.10.13	255.255.255.0	192.168.10.1

3. /27 Subnetting on 192.168.1.0

192.168.1.0 is a Class C network (default /24). /27 means 3 extra network bits, leaving 5 host bits.

Subnet mask = 255.255.255.224, block size = $256 - 224 = 32$.

Valid subnets = $2^3 = 8$

Hosts per subnet = $2^5 - 2 = 30$

Subnet	Network Address	Usable Host Range	Broadcast Address
Subnet 1	192.168.1.0	192.168.1.1 - 192.168.1.30	192.168.1.31
Subnet 2	192.168.1.32	192.168.1.33 - 192.168.1.62	192.168.1.63
Subnet 3	192.168.1.64	192.168.1.65 - 192.168.1.94	192.168.1.95

Usable host IPs: Subnet 1 = 192.168.1.1 to 192.168.1.30; Subnet 2 = 192.168.1.33 to 192.168.1.62; Subnet 3 = 192.168.1.65 to 192.168.1.94 (30 hosts each).

4. IP 10.0.5.17 with Mask 255.255.255.0

Mask 255.255.255.0 = 24 network bits = /24.

Bitwise AND:

IP : 00001010.00000000.00000101.00010001 (10.0.5.17)

Mask : 11111111.11111111.11111111.00000000 (255.255.255.0)

Result: 00001010.00000000.00000101.00000000 = 10.0.5.0

Item	Value
CIDR notation	10.0.5.17/24
Network address	10.0.5.0
Broadcast address	10.0.5.255 (host bits all 1)
Valid host range	10.0.5.1 to 10.0.5.254
Total usable hosts	$2^8 - 2 = 254$

5. Verification with Subnet Calculator

Enter 10.0.5.17 with mask 255.255.255.0 (/24) at <https://www.subnet-calculator.com/>. The tool should show:

Field	Expected Tool Output
Network Address	10.0.5.0
Broadcast Address	10.0.5.255
Usable Host Range	10.0.5.1 - 10.0.5.254
Usable Hosts	254

Screenshot: [Paste your own calculator screenshot here]

One-line conclusion: My manual answer matched the tool's output exactly for network address, broadcast address and host range.